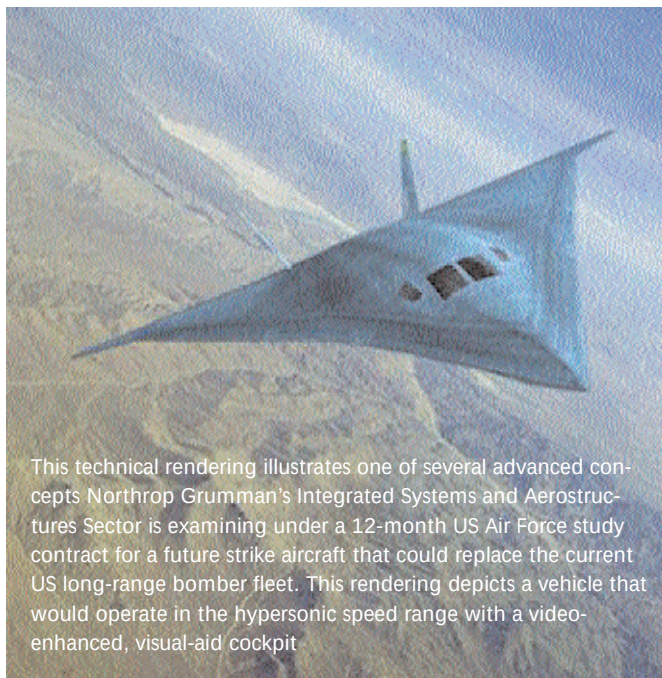




This technical rendering illustrates one of several advanced concepts Northrop Grumman's Integrated Systems and Aerostructures Sector is examining under a 12-month US Air Force study contract for a future strike aircraft that could replace the current US long-range bomber fleet. This rendering depicts a vehicle that would operate in the hypersonic speed range with a video-enhanced, visual-aid cockpit



Global Hawk is a high-altitude, long-endurance, unmanned aerial reconnaissance system designed to provide military field commanders with high-resolution, near real-time imagery of large geographic areas

which is a strategic cruise missile with range estimated at up to 1,500 miles.

Given its radar-evading ability, the Stealth fighter has been used in attacks deep behind enemy lines. It is said that the first Iraqi anti-aircraft weapons were fired after bombs from the F-117s hit their targets with an astounding rate of success. F-117s were the first aircrafts to strike Baghdad in the opening minutes of the Gulf war. Footage from onboard cameras shows a number of 200-pound bombs from F-117s scoring direct hits on Iraqi strategic targets and mobile missile launchers. Air Force commanders reported that the F-117 flew 3 per cent of total missions but destroyed over 43 per cent of total targets.

//

Be subtle! Be subtle! And use your spies for every kind of business //

The art of war

Robot planes called Predators circled slowly in the skies above Afghanistan, transmitting live images of Taliban troops with no risk to the pilots. Predators are capable of staying aloft for 40 hours, lingering over no-man's land, snooping on a single target or a hundred of them and transmitting back vivid, live pictures. This plane is a sign of things to come, when troops made of metal and electronics do the fighting, not humans. It's only a matter of time before fighter planes without any pilots joust in some robot dogfight in the sky.

iRobot of Somerville, Massachusetts, is a small machine modelled on the crab, which is devoted to minesweeping. It can be used in amphibious assaults as it has the ability to navigate difficult terrain, attach itself to a mine, and detonate itself. To imitate an ant farm, iRobot is building a swarm of tiny robots called 'joeys'. Possessing very simple intelligence, the joeys are meant to work together to explore a battlefield or disaster area. They communicate by 'pheromones' which are transmitted by infrared beacons, and each joey has a small tank of water, called a bladder, from which it dribbles a trail that

shows other joeys where it has gone. Like ants, the joeys know little of their universe. But when they find something interesting, they tell others around them.

The most incredible statistic of the Afghanistan war was that the first American soldier died after three months of combat! Thanks to the Predator spy plane and other unmanned drones and gunships along with satellites and thermal imaging devices, battlefield robots can do all the dirty work in future wars—from minesweeping to reconnaissance, from killing to guarding. They would be exposed in risky situations, and would be expendable.

The challenge lies in designing machines with enough intelligence and mobility to handle the unpredictability of the real world. At Sandia Lab in New Mexico, scientists are at work to start a robotics revolution by 2020, with advances in biometrics, swarm intelligence and AI.

Such advances could further divide the gap between warring factions. As it is, the actual battles today are not decisive or heroic anymore, and a proxy war through machines would make war even more confusing, distant, and one-sided. The actual battles, with its paradoxes, and infomania, would seem best described as postmodern. ■

SRIRAM SHARMA



With the thermal sight at night or video sight during the day, the 21st Century Land Warrior can shoot with great accuracy while exposing only his hands